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BACHELOR'S THESIS

Research Project

LegalBechRu: A Benchmark for evaluating Retrieval-Augmented Generation Systems in context of Russian Legislation

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Abstact

Retrieval augmented Generation systems have become increasingly common in our lives and, among other things, demonstrate great potential for legal topics using artificial intelligence. Unfortunately, this is not very common when working with Russian legal materials. There are a number of tests that already check how large language models generate legal content, but there still remains a significant gap in the assessment of the search component in relation specifically to Russian legislation.

LegalBenchRu-RAG satisfies this need as the first stage and benchmark designed to evaluate the effectiveness of RAG systems for extracting information from Russian legal documents. The methodology based on LegalBench-Rag combines well-established approaches and adapts to unique challenges and examples emerging in Russian legal texts.

Accuracy in this matter is quite important. LegalBenchRu-Rag focuses on extracting only the most significant fragments from Russian legal documents, instead of extracting entire documents and extensive text fragments. This method makes it possible to avoid problems that may occur in context-overloaded windows by reducing processing costs, reducing latency, and preventing AI agents from forgetting key information and starting to reproduce hallucinations.

Russian legal documents are not the easiest to work with, in part this has created difficulties in development and will be a key issue in terms of further work related to this topic. All this is due to the citation style, specialized terminology, and overall structure of all documentation in the Russian legal corpus. The process of creating a reference text included comparing queries with sources in Russian legal texts, without forgetting about citations, for further work after providing a reference response. LegalBenchRu-Rag serves as a practical tool for organizations and researchers involved in improving RAG systems for Russian legal applications. The benchmark is aimed at the development of the industry and the development of real legal work in the Russian conditions.

Introduction

In the modern world, Ai agents occupy a fairly significant part. Most often, such agents are used locally, for any tasks of a particular business, or purely for personal purposes. In many countries, AI agents are gradually being introduced into broader areas as second opinions or advisors, where I can listen to their opinions. All this is due to the fact that almost all data is publicly available on the Internet, and agents just have access to this information and they can quickly find a suitable or, as it seems to us, a suitable answer. Artificial intelligence (AI) and advances in natural language processing (NLP) open up new possibilities for automating a multitude of tasks. In our case, we will consider the automation of legal processes. One of the most promising technologies is Retrieval Augmented Generation (RAG), which is promising due to its mechanism of operation - searching for relevant information in knowledge bases and generating responses based on language models. However, the effectiveness of using RAG systems in the legal field is a slightly different matter, since it requires an accurate assessment of their ability to work with specialized texts, documents, and regulations that have such important aspects as referencing from one code to another, quoting various court decisions, and even the overall structure of this documentation. it is not obvious to both ordinary users and AI agents.

It is worth noting that there are already international benchmarks, such as LegalBench, for English-speaking legal systems. Unfortunately, there are no similar approaches for evaluating the effectiveness of RAG systems in the context of Russian legislation. This creates a number of difficulties that, in their own way, slow down the process of developing AI agents under Russian law, simply because there is no test that would assess the effectiveness of the work. This work serves as an example of how to learn from the experience of foreign colleagues and project it onto Russian law, which has unique features, including a complex hierarchy of regulations, specific terminology, and frequent changes in legislation.

The relevance of the research is due to the growing demand for AI solutions, which should by no means replace lawyers who are familiar to everyone, on the contrary, this approach should simplify their work with routine tasks and allow both the public to receive quick and accurate answers to their legal questions, and qualified lawyers to pay more attention not just to finding information. namely, the process of finding solutions for each specific case. The developed benchmark will make it possible to compare RAG systems, identify their weaknesses and improve algorithms for working with Russian-language legal texts.

The scientific novelty of the work lies in the creation of the first open test suite for evaluating RAG systems in the context of Russian legislation, as well as in the proposal of a methodology for adapting such solutions to the legal field. The practical significance lies in the fact that the research results can be used by developers of legal AI services, government agencies and law firms to improve the efficiency of working with legal information.

Basic terms and definitions

Retrieval-Augmented Generation (RAG) – A hybrid AI system that combines retrieval of information in a knowledge base and generation of responses using a language model.

Large Language Model (LLM) – A large language model (for example, GPT-4) capable of processing and generating text.

AI Agent – An autonomous program based on And that performs tasks

Benchmark – A standardized test for evaluating the performance of algorithms

Dataset – Structured data set

Legal Corpus – Collection of legal documents

Citation – A link to the exact document fragment (a key requirement in legal rags)

Domain-Specific Nuances – Unique features of legal texts (length, terminology, structure)

Ground Truth – Reference data marked up by people

Natural Language Processing (NLP) – AI techniques for analyzing and generating legal text

Multimodal Legal AI – Combining text, voice, and visual data (e.g., analyzing court recordings).

Explainable AI (XAI) – Making AI's legal reasoning transparent.

AI as a Legal "Co-Pilot" – Augmenting (not replacing) lawyers' workflows.

Related works

Retrieval Augmented Generation (RAG)

Retrieval-Augmented Generation (RAG) systems are intelligent generative models that work with external knowledge bases. As described in the work of Lewis and co-authors (2020), such a system operates with a specially organized collection of documents D, where each individual document D represents a text record of legal content.

In a typical implementation of legal document processing, the system first segments each document di into logical fragments $c_j \in C_i$. Then each received fragment is transformed into a vector representation using a special vector coding model. After completing the document indexing procedure, the user can formulate a text query q, which is similarly transformed into a vector space.

At the next stage, the system identifies the k most relevant fragments $R_q = \{r_1, r_2, ..., r_K\}$ using vector similarity metrics such as cosine proximity. The final answer is generated by a large language model (LLM) based on:

1. the extracted text fragments
2. the original user query
3. the system error.

This whole process can be formally represented by the following work scheme:

$$\text{Contextual Retriever}(q, D) \rightarrow R_q$$

$$LLM_p(q, R_q) \rightarrow \text{answer}$$

Contextual Retriever: The core function of this component involves identifying and extracting relevant data from an external knowledge repository, followed by compiling a set of contextually significant fragments Rq. A typical RAG architecture implementation employs a dual-encoder retrieval model (bi-encoder), such as DPR (Karpukhin et al., 2020), which demonstrates high efficiency in processing information queries.

The retrieval process usually includes a results re-ranking phase. In the first step, the system selects the top k′ most relevant items using the bi-encoder. Then, each of these items is analyzed by a cross-encoder model that calculates semantic similarity metrics between the query q and each fragment. Based on this analysis, the top k most relevant results are returned, where $k < k'$.

Answer Generation Module: This component, based on a sequence-to-sequence (seq 2 seq) architecture, processes the original query in combination with the retrieved contextual fragments. The final output is an answer $y_{j,q}$ with the model estimating the probability $P_G(y_{j,q} | q, r_{j,q})$ of its correctness relative to the given query and provided context.

Retrieval Augmented Generation (RAG) benchmarks

The development of question-answering evaluation frameworks predates the emergence of RAG architectures, with established datasets like HotPotQA (Yang et al., 2018) remaining relevant in current research. In recent years, specialized evaluation tools have emerged specifically designed to measure RAG system performance. Notable examples include the Retrieval Augmented Generation Benchmark (RGB) developed by Chen et al. (2023) and the RECALL framework proposed by Liu et al. (2023). These evaluation methods primarily focus on straightforward scenarios where responses can be generated through single-step retrieval from general knowledge sources.

To address more sophisticated requirements, newer benchmarks like MultiHop-RAG (Tang and Yang, 2024) have been developed. These advanced frameworks test language models' ability to perform iterative information retrieval and reasoning across multiple knowledge sources, representing a significant evolution in evaluation methodology for complex query processing.

LegalBench

The LegalBench framework (Guha et al., 2023) represents a significant advancement in evaluating large language models' capacity for legal analysis. This comprehensive assessment tool comprises 162 distinct evaluation scenarios spanning six fundamental categories of jurisprudential logic. What distinguishes this benchmark is its collaborative development methodology, which actively incorporated contributions from practicing legal experts throughout its creation. The direct involvement of legal professionals in designing evaluation criteria ensures that the benchmark assesses either practically relevant analytical skills or intellectually compelling aspects of legal reasoning that practitioners value.

LegalBench-RAG

LegalBench-RAG – the first specialized benchmark for assessing the retrieval component of RAG systems in legal applications. While existing benchmarks like Legal Bench evaluate LLMs' generative capabilities, they overlook critical retrieval performance metrics. LegalBench-RAG addresses this gap by focusing on precise extraction of legally relevant text segments rather than entire documents, minimizing hallucination risks and optimizing context window usage.

Problem definition

There is an urgent need for specialized tools to evaluate the effectiveness of Retrieval-Augmented Generation (RAG) systems in relation to Russian legislation. Unlike existing benchmarks such as LegalBench, which are focused on English-language legal systems, Russian legislation has its own unique features that require a special approach. The main problem is that Russian legislation has a rather complex hierarchical structure, which includes many interconnected codes, federal laws, resolutions, acts and other regulations. All these legal documents are related in one way or another.

The difficulty also lies in the Russian language itself and the legal terminology, which differs significantly from the generally accepted language, English, and requires a deeper understanding of the context.In addition, Russian legislation is subject to frequent changes and additions, which requires RAG systems to constantly adapt and update their knowledge.

Due to the fact that there are no standardized criteria in Russian legislation, this makes it difficult to develop and evaluate AI agents designed to automate legal processes. All this hinders the introduction of AI solutions into legal practice in Russia, reduces the efficiency of lawyers and makes it difficult for citizens to quickly answer various questions regarding legal information. Without such a benchmark, it is impossible to objectively assess the ability of RAG systems to extract relevant fragments from Russian legal documents, take into account their hierarchical structure and specific terminology, and effectively cope with frequent changes in legislation. This leads to the risk of errors and false information, which is unacceptable in the legal field.

A Benchmarking Dataset: LegalBenchRu-RAG

LegalBenchRu-RAG construction

In this section, we provide detailed information about building the LegalBenchRu-RAG dataset. Let's see what regulatory legal acts of Russian legislation this dataset consists of.

Support for research

Our development is based on LegalBecnh, a legal reasoning test developed jointly from 162 tasks covering six different types of legal reasoning. LegalBench is a widely known dataset that tests the understanding of legal concepts and their legal reasoning by providing them with the precise context they need to answer legal questions. A key role in creating your own dataset was played by the fact that the LegalBench dataset is based on English and foreign law. Unfortunately, we have not found analogues of such datasets in Russia, which significantly complicates the development of such a solution. In the absence of specific analogues, it is quite difficult to create something from scratch when there is nothing to rely on. In Russia, the main focus is on analyzing legal documents, rather than building Q&A datasets to obtain information on law.

Dataset sources

There are open resources such as Consultant Plus, Garant, and Codex. All information with all laws, codes, acts and other legal documents is stored on these resources. To collect the LegalBenchRu dataset, we used 10 different legal documents, namely:

The Civil Code of the Russian Federation

The Civil Procedure Code of the Russian Federation

Urban Planning Code of the Russian Federation

Housing Code

The Land Code

The Administrative Code

The Tax Code

The Labor Code

The Criminal Code

Federal Law on Advertising

Consumer Protection Law

Starting Point: LegalBenchRu

For the first stage of work, the full original files of all the above documents were downloaded from the Garant website for further work with them. There were several options for how to create this dataset. One of the options was to involve experienced lawyers for a long and painstaking work with the preparation of questions and the subsequent giving of detailed and complete answers to them, but this approach seemed quite long and quite expensive, requiring several other resources than allocated for writing this thesis. Another approach is to compile detailed prompta for large neural networks that are able to formulate appropriate questions and provide clear, detailed answers using small knowledge bases with legal documents. In fact, this is an approach similar to RAG systems, we have large language models and a knowledge base (a file that we upload to them). With the help of clear promptness, we tried to avoid large hallucinations associated with free Internet access.

The way to compose questions and answers

To generate questions and answers, we used several publicly available services, from OpenAI Chat-GPT, DeepSeek and Gemini, all of the above listed on the latest available versions published for widespread use. The results were different in one way or another. For example, using Gemini, the neural network not only provided answers to the questions asked using the provided files, but also outlined recommendations and schemes for solving certain problems each time. It seemed to us that for a dataset of question-answer pairs, such answers would not be entirely satisfactory, since the answers to the "reference" questions should be as substantive as possible and without unnecessary water, since a benchmark would later be built on this dataset to evaluate other AI agents in the legal field, who simply by virtue of their databases They may not have enough information to provide recommendations in addition to answering questions.

Synthetic or as close as possible to real questions

It was important to take into account that the questions were not just "dry", as they are usually generated by various language models, because they still differ from people in the algorithm of their thinking, but to make them exactly "alive", as similar as possible to those that people would actually ask. To do this, we used the most popular browser in Russia, Yandex, with which we found the most frequently asked questions from ordinary people on topics related to our previously selected documents, and after long iterations and gradual changes to the software, we came to generate the most human-friendly questions from neural networks. Most of the questions in the dataset are close to those that real people could ask. This is done to ensure that our dataset is close to the conditions and formulations that may actually come from users. All this made it possible to make the questions as non-synthetic as possible for better subsequent results and greater convenience for further improvement and improvement of our dataset.

Quoting the original legal documentation

It was also important to take into account the fact that in most cases we should refer to the number of the article or the name of the legal act by which this or that answer was given. This is done in order to independently verify that the answers received to the questions asked really coincide with the topic. Often, in the generation of responses, one could find that the question was asked with a focus on a particular article, and the answer was given with an emphasis on a completely different one. In the end, we managed to ensure that the responses received contained references to paragraphs, article numbers, or at least just to legal documents that were valid on the issue.

For all legal documents, namely:

The Civil Code of the Russian Federation

The Civil Procedure Code of the Russian Federation

Urban Planning Code of the Russian Federation

Housing Code

The Land Code

The Administrative Code

The Tax Code

The Labor Code

The Criminal Code

Federal Law on Advertising

Consumer Protection Law

Number of question-answer pairs in the dataset

From 50 to 150 pairs of question-answers were compiled. The choice of such a quantity is quite easy to explain. For example, only 50 pairs of question-answers were generated for the Federal Law on Advertising, because the document itself is quite compact and it is not advisable to compose a question for each of its articles, because part of the paragraph is not even such that you can ask any question, they are informative in nature of the law itself. For example, 150 question-answer pairs were compiled for the Criminal Code of the Russian Federation, because currently there are 360 articles in this code, 104 of them in the General part, and 256 in the Special part. Articles of the Criminal Code are numbered in Arabic numerals. To be as close as possible to the questions that, in theory, can come from users, you need to cover as many articles as possible, so the number of question-answer pairs has been increased. If the codes were approximately the same size, then they were taken as the standard number - the number of 100 pairs of question-answers.

Assembling a dataset

After successfully generating all the question-answer pairs for the above legal documents, we have compiled them individually into .txt files. After that, each file was converted to .cvs and .xlsx format. So we got 11 files with question-answer pairs. Then we combined all these files into 1 single dataset of 1000 lines. This is how we got the LegalBenchRu dataset.

We performed several operations to see what our assembled dataset looks like. We have calculated the average length of the question, which is 10.23 words, which tells us that the questions are of average length and are not very simple, nor do they contain any complex constructions. We also found the average length of the answers, which was 31.50 words, from which we can conclude that the answers to the questions posed are not monosyllabic, but they are not overloaded with complex constructions and they practically do not provide recommendations, but only the answers to the questions themselves and mentions of legal documents that the agent relies on when answering..

Possible future of dataset development

Suppose a logical question might follow, why didn't we take, for example, 300 or 400 pairs of question answers, or at least, using the example of the Criminal Code, where there are 360 articles, why did we take only 150. The answer is quite simple, the ratio of question-answer pairs for each legal document we choose. We felt that it would be incorrect to use such a large gap between legal documents and ask 50 questions about some things and 250 questions about others. Too much range of different topics would most likely have affected the final result, which we certainly did not want. To use a large number of question-answer pairs more correctly in the future, it is more logical to set the same number with some small error for each selected document. Indeed, you can come up with and generate more than one question per article, but for example, 10 or 20 different questions can be formulated for one article, but in our work we did not try to create some kind of ideal dataset, we just want to lay the foundation for work in this direction, so that later a much larger community can conduct all work on improving the quality of the benchmark for evaluating the quality of responses provided using AI agents on legal topics in the Russian Federation

Limitations

We acknowledge several limitations inherent in the LegalBenchRu-RAG dataset. Primarily, the question-answer pairs were generated using large language models rather than being authored by human legal experts, introducing a potential risk of factual inaccuracies, "hallucinations," or a lack of subtle legal nuance. Compounding this, due to resource constraints, the dataset did not undergo systematic review or validation by qualified Russian legal professionals. Although inspired by real user search trends, the queries remain synthetically generated by AI and may not fully replicate the exact phrasing or intent of genuine user questions. Furthermore, the dataset's scope is limited; while based on 11 significant Russian legal documents, this selection is not exhaustive, and the depth of coverage varies across these documents. The benchmark is also intentionally designed to assess the retrieval of specific text snippets and citations, deliberately excluding the evaluation of more complex tasks like multi-document reasoning or detailed legal analysis, which were filtered out during generation. Finally, while efforts were made to ensure accurate citations, the AI generation process presented challenges, meaning some residual errors, imprecise references, or missing citations might persist.

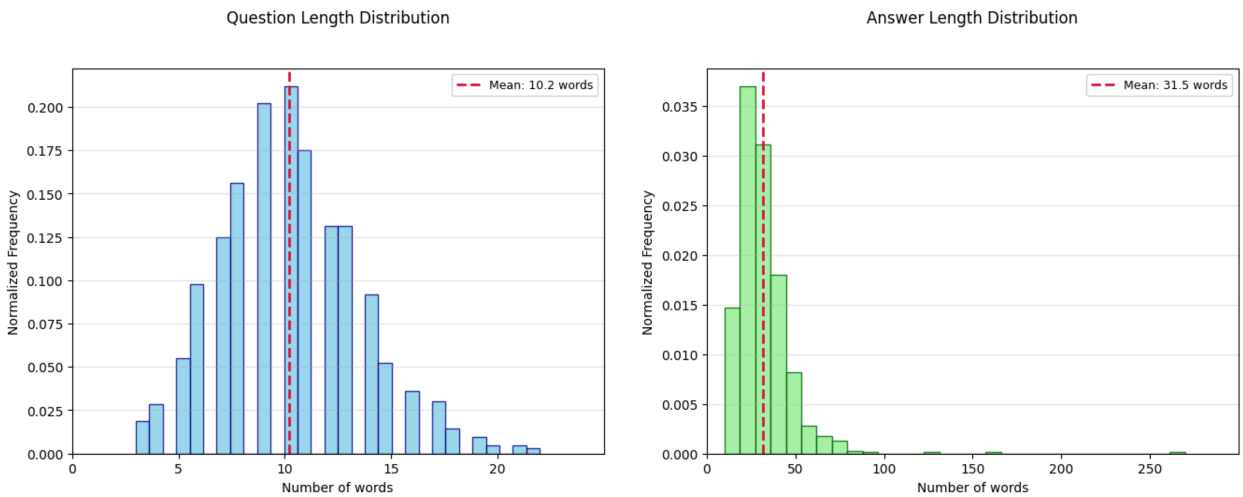

Table 1.
Question & Answer Length Distribution

Dataset for the RAG system benchmark

The created dataset of 1000 lines is suitable for use on large language models without using RAG systems. However, it is not ideal for use with RAG systems, because it consists of only 2 columns - questions and answers. That's why we decided to create and mark up another dataset from existing data. We took 8 out of 11 legal documents, separated 50 pairs of question-answers from each of them and added several more columns that contain the following information: the document on which the question was asked and the answer was generated (source file), the beginning of the symbol, because we used legal documents for the answers to reduce the number of possible hallucinations, the end of a character from the same document and the extracted snippet itself. All this is necessary to build a more accurate benchmark for evaluating the accuracy of AI agents in generating answers to legal questions under Russian law using RAG systems. As a result, we got another dataset consisting of 420 pairs of question-answers and additional information for each pair.

Question distribution by sources

- Градостроительный кодекс Российской Федерации от 29 декабря 2004 г. N 190-ФЗ (с изменениями и дополнениями).txt: 15.9% of questions
- Трудовой кодекс Российской Федерации от 30 декабря 2001 г. N 197-ФЗ (ТК РФ) (с изменениями и дополнениями).txt: 13.3% of questions
- Жилищный кодекс РФ.txt: 12.5% of questions
- Гражданский процессуальный кодекс Российской Федерации от 14 ноября 2002 г. N 138-ФЗ (ГПК РФ) (с изменениями и дополнениями).txt: 12.0% of questions
- Закон РФ от 7 февраля 1992 г N 2300 I О защите прав потребителей с изменениями.txt: 12.0% of questions
- Федеральный закон от 13 марта 2006 г N 38 ФЗ О рекламе с изменениями.txt: 12.0% of questions
- Земельный кодекс РФ.txt: 11.8% of questions
- Уголовный кодекс Российской Федерации от 13 июня 1996 г. N 63-ФЗ (УК РФ) (с изменениями и дополнениями).txt: 10.4% of questions

Distribution analysis

1. Total source categories: 8
2. Largest share: Градостроительный кодекс Российской Федерации от 29 декабря 2004 г. N 190-ФЗ (с изменениями и дополнениями).txt (15.9%)
3. Smallest share: Уголовный кодекс Российской Федерации от 13 июня 1996 г. N 63-ФЗ (УК РФ) (с изменениями и дополнениями).txt (10.4%)
4. Largest/smallest share ratio: 1.5:1

Key observations:

- Градостроительный кодекс Российской Федерации от 29 декабря 2004 г. N 190-ФЗ (с изменениями и дополнениями).txt dominates with 15.9% of all questions
- Трудовой кодекс Российской Федерации от 30 декабря 2001 г. N 197-ФЗ (ТК РФ) (с изменениями и дополнениями).txt and Жилищный кодекс РФ.txt together account for 25.8%
- Distribution shows balance toward Градостроительный кодекс Российской Федерации от 29 декабря 2004 г

Additional statistics:

Total questions: 419

Average questions per source: 52.4

Median share: 12.0

RAG-system using LegalBenchRu-RAG

The development of a benchmark for testing the Retrieval-Augmented Generation (RAG) system became a key part of my graduation thesis. The task was to create a universal tool that would allow comparing different machine learning models on legal issues related to Russian legislation. The code was organized in a sequence of blocks, each of which solved its own task: from data preparation to evaluation of results. I paid special attention to the retriever, as its quality directly affected the final responses. After evaluating the retriever, I got the metrics: Precision@5 was 0.1517, and Recall@5 was 0.6393. These indicators have become the starting point for analysis and improvements. It is important to note that there are practically no such data centers in Russia focused on legal issues with reference to regulatory acts, which makes my work unique.

The dataset We used was a CSV file with five columns: Question, Answer, Source file, Character Start, Character End, and Extracted snippet. It contained examples of legal issues, correct answers, and relevant text fragments from regulations. Creating such a dataset has become one of the most difficult tasks, as there are no ready-made datasets in Russia that combine questions and excerpts from legislation. We had to manually collect the questions and search for suitable articles, which took a lot of time. As a result, we limited myself to 50 test entries to speed up debugging.

The building was formed from PDF files, such as the Urban Planning Code. We chose PDF because it is the format in which official documents are most often published. Text extraction proved difficult: tables, footnotes, and line breaks created unnecessary noise. We used the pdfplumber library, which allowed me to extract text page by page, but we still had to clean up the data, removing empty lines and artifacts. To make the texts easier to process, we divided them into sentences, limiting the fragments to 500 words so as not to lose their meaning.

We decided to write a retriever because the quality of the responses depended on its ability to find the correct text fragments. We used the paraphrase-multilingual-MiniLM-L12-v2 model to create embeddings that converted texts into numeric vectors. These vectors were saved to the index using the FAISS library to quickly find the closest fragments to the cosine similarity question. To increase accuracy, we added filtering, which selected only those fragments that contained enough words from the question.

After the first implementation, I evaluated the retriever by calculating the metrics Precision@k and Recall@5. Precision@5 was 0.1517, which meant that only in 15% of cases the desired fragment was in the top 5. Recall@K was higher — 0.6393, that is, in 64% of cases, the correct fragment was still among the selected ones. These figures showed that the retriever lacks accuracy, especially for complex questions, such as disputes with neighbors. We assumed that the problem might be in the embedding model, but there are still significantly more models for English than for Russian and they are developing much faster. It may be necessary to use large multilingual models, access to which is paid, which does not really fit into this thesis, or they have quite a lot of weight and, again, will not suit us due to the timing of this work.

If we compare the performance of LegalBenchRag Precision at 15%, it looks very good. Unfortunately, due to the lack of such datasets in Russian, we have nothing to compare them with, but in theory we can compare them with a foreign article that described 4 different datasets and evaluated Precision and Recall. In our opinion, the result for Precision of 15% is high, because on average Precision was in the range from 5 to 6, and only one dataset was jn 3 to 16. As for Recall, the situation here is not so effective, we did not manage to increase this metric much compared to foreign datasets, but the result of 65% is very good, on average for other datasets it ranged from 10 to 40%, but there were also indicators of 70-85%. Based on all of the above, we can conclude that the dataset is well-designed, as is the corpus of documents on which we created it.

8 legal documents in Russian were used to create the LegalBenchRu-RAG dataset. Due to the fact that there are no similar datasets in Russia, this one will serve as an example for further work and development.

Results and discussions

As part of our thesis, we built the LegalBenchRu-RAG benchmark to assess Retrieval-Augmented Generation (RAG) systems for Russian legal texts. We created a dataset with 420 question-answer pairs, sourced from eight key legal documents like the Urban Planning Code, Civil Code, and Labor Code. For each pair, we included details such as the source document, character positions, and the extracted snippet, which allowed us to thoroughly test the system's retrieval accuracy.

Our main focus was on the retriever, since it plays a crucial role in finding the right legal snippets. We used the paraphrase-multilingual-MiniLM-L12-v2 model to create embeddings and FAISS to index them for fast similarity searches. After running our tests, we calculated the retriever's performance: Precision@k came out at 0.1517, meaning that in 15% of cases, the correct snippet was among the top five retrieved results. Recall@k was **0.**6393, showing that in 64% of cases, the relevant snippet was included in the selection. We felt these numbers were a good starting point. The recall was encouraging—it meant our retriever often found the right fragments, which was important for a dataset dealing with Russian legal texts. The precision, though lower, was in line with what we'd seen in foreign studies, where precision often ranged between 5% and 16%. Given the tricky nature of Russian legal language, with its complex terms and structure, we thought this was a solid result for our first attempt.

What makes LegalBenchRu-RAG stand out is its focus on Russian legislation—an area where no similar datasets exist. In Russia, there's nothing like the English-language LegalBench, which has plenty of resources to draw from. We had to start from scratch, and creating a benchmark that covers real-world legal topics, from urban planning to consumer rights, felt like a big achievement. The dataset's question distribution also showed balance: 15.9% of questions came from the Urban Planning Code, 10.4% from the Criminal Code, and the rest spread across other sources. Questions averaged 10.23 words, which we thought reflected the kind of concise queries people might ask, while answers averaged 31.50 words, giving clear references to legal articles.

Conclusion

Why LegalBenchRu-RAG is an important step for the development of AI in Russian law

Creating legal datasets is not an easy task. This requires not only deep knowledge of law, but also a significant investment of time and resources. That is why the appearance of LegalBenchRu-RAG is an event worthy of attention.

Filling the gap in the Russian legal IT sphere

Up to this point, there was no open tool specifically designed to check how well artificial intelligence copes with extracting information from Russian legal documents. Most of the similar solutions are focused on the English-speaking legal system. LegalBenchRu-RAG fills this gap, taking into account the peculiarities of the structure, language and terminology of Russian legislation.

Making legal AI development more accessible

Usually, creating large legal datasets is expensive and requires the participation of entire teams of lawyers. We approached it differently. Based on real questions from Yandex users and using modern language models (such as ChatGPT, Gemini and DeepSeek), we managed to create a thousand question—answer pairs. This approach has significantly reduced costs and made the development of legal AI systems accessible to a larger number of teams in Russia.

How can I use LegalBenchRu-RAG?

The main value of LegalBenchRu-RAG is that it makes it possible to objectively compare how different AI systems cope with the search for information in legal texts. Can the model find the right article in the Civil Code? Does she understand how one law refers to another? The questions in the dataset are selected to reflect real user requests, for example: "What will happen for non-payment of taxes?" Moreover, each response contains a link to a specific article of the law, which makes it easy to verify the correctness of the answer.

Contribution to science and future development

The initial version of the dataset covers important areas such as tax and civil law. But the structure of the project makes it easy to expand it: add other codes, increase the number of questions, and adapt it to new tasks. At the same time, the emphasis is not on complex legal analysis, but on accurately extracting the necessary information from the text. This approach helps to avoid serious mistakes when using AI in legally significant situations.

Benefits for the economy and society

LegalBenchRu-RAG simplifies entry into the field of legal AI for startups, research groups and developers in Russia. This drives innovation. In the long term, such technologies will be able to provide citizens with understandable and reasonable legal answers, referring to the current legislation. This increases the level of trust in digital services and makes legal information more accessible to everyone.

Looking back on our project, we're proud to have created LegalBenchRu-RAG—the first open benchmark for testing RAG systems on Russian legislation. In Russia, there aren't any datasets that combine legal questions with snippets from laws, so building this from the ground up was a challenge for us. We had to collect legal texts, write questions that sounded natural, and make sure answers pointed to the right articles. The result is a tool that lets developers test how well their AI can handle Russian legal documents, which we think is a step forward for legal tech in our country.

The retriever's results gave us a lot to think about: Precision@k at 0.1517 and Recall@k at 0.6393. These numbers showed us that our system could often find the right snippets, with recall being particularly strong at 64%. The precision, at 15%, matched what we'd seen in some international benchmarks, which typically range from 5% to 16%. For Russian legislation, with its unique structure and language, we felt this was a promising start. It gave us a baseline to build on, whether by tweaking the retriever or adding more data down the line.

LegalBenchRu-RAG fills a real gap in Russia. Unlike English-speaking countries with benchmarks like LegalBench, we didn't have anything to work with here. Our dataset, covering topics from urban planning to criminal law, lets developers test their systems on questions like "What are the rules for building permits?"—the kind of thing people actually ask. By making this benchmark open, we hope it encourages more people to get involved, helping build AI tools that make legal information easier to access for everyone in Russia.

List of resources:

- Vladimir Karpukhin, Barlas Oguz, Sewon Min, Patrick ˘ Lewis, Ledell Wu, Sergey Edunov, Danqi Chen, and Wen tau Yih. Dense passage retrieval for open-domain question answering, 2020. URL https://arxiv.org/ abs/2004.04906

- Patrick S. H. Lewis, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, Heinrich Kuttler, Mike Lewis, Wen-tau Yih, Tim Rocktaschel, Sebastian Riedel, and Douwe Kiela. Retrieval-augmented generation for knowledge-intensive NLP tasks. CoRR, abs/2005.11401, 2020. URL https://arxiv.org/ abs/2005.11401.

- Yixuan Tang and Yi Yang. Multihop-rag: Benchmarking retrieval-augmented generation for multi-hop queries, 2024. URL https://arxiv.org/abs/ 2401.15391.

- Zhilin Yang, Peng Qi, Saizheng Zhang, Yoshua Bengio, William W. Cohen, Ruslan Salakhutdinov, and Christopher D. Manning. Hotpotqa: A dataset for diverse, explainable multi-hop question answering, 2018. URL https://arxiv.org/abs/1809.09600.

- Nicholas Pipitone, Ghita Houir Alami. LegalBench-RAG: A Benchmark for Retrieval-Augmented Generation in the Legal Domain. URL https://arxiv.org/html/2408.10343v1

- Lewis, P., Oguz, B., Rinott, R., Riedel, S., & Stoyanov, V. (2020). Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks. URL https://arxiv.org/pdf/2005.11401

- Neel Guha, Julian Nyarko, Daniel E. Ho, Christopher Re,´ Adam Chilton, Aditya Narayana, Alex Chohlas-Wood, Austin Peters, Brandon Waldon, Daniel N. Rockmore, Diego Zambrano, Dmitry Talisman, Enam Hoque, Faiz Surani, Frank Fagan, Galit Sarfaty, Gregory M. Dickinson, Haggai Porat, Jason Hegland, Jessica Wu, Joe Nudell, Joel Niklaus, John Nay, Jonathan H. Choi, Kevin 9 LegalBench-RAG: A Benchmark for Retrieval-Augmented Generation in the Legal Domain Tobia, Margaret Hagan, Megan Ma, Michael Livermore, Nikon Rasumov-Rahe, Nils Holzenberger, Noam Kolt, Peter Henderson, Sean Rehaag, Sharad Goel, Shang Gao, Spencer Williams, Sunny Gandhi, Tom Zur, Varun Iyer, and Zehua Li. Legalbench: A collaboratively built benchmark for measuring legal reasoning in large language models, 2023. URL https://arxiv.org/abs/ 2308.11462.

- Chen, W., Jin, Y., Wang, H., Zhao, W. X., & Wen, J. R. (2023). RGB: A Benchmark for Retrieval-Augmented Generation. URL https://arxiv.org/html/2309.01431v2

- Liu, Y., Wang, Y., Zhang, X., Zhang, S., & Yin, D. (2023). RECALL: Retrieval-Augmented Generation Evaluation with Document-Level Annotations. URL https://arxiv.org/pdf/2312.10466

- Github: https://github.com/AndreisLavrov/LegalBenchRu-RAG

Legal documents used to create the dataset (downloaded from the Garant website):

- Гражданский кодекс Российской Федерации
- Гражданский процессуальный кодекс Российской Федерации
- Градостроительный кодекс Российской Федерации
- Жилищный кодекс Российской Федерации
- Земельный кодекс Российской Федерации
- Кодекс Российской Федерации об административных правонарушениях
- Налоговый кодекс Российской Федерации
- Трудовой кодекс Российской Федерации
- Уголовный кодекс Российской Федерации
- Федеральный закон «О рекламе» от 13 марта 2006 г. № 38-ФЗ
- Закон РФ «О защите прав потребителей» от 7 февраля 1992 г. № 2300-1